

Invasive Plant Survey - Belt Transect Protocol

This protocol will be followed for surveying invasive plants near trails in Reno, Nevada. Each student group will survey the presence and absence of 23 species invasive to the Great Basin at a sampling site location using the *belt transect* method. At each sampling site location, 3 survey collection points will be selected along a trail by the student group, which will serve as the starting points for the 3 *belt transects*. Belt transects will be 30 meters in length and 6 meters in width, divided into 1 meter by 1 meter sampling squares (see *figure 1*). Students will use pacing to measure the 30 meter length of the transect and a tape measure to mark the width. Students will be provided a plant ID list to help identify plant species that are present in each sampling square, and a data sheet will be used to record the data.

1. Students will receive a sampling site location, and start and end points within the site, which will determine the boundaries for the 3 belt transects.
2. Choose 3 survey collection points on the trail. The 3 survey collection points have to be within the boundaries of the start and end points.
3. Measure 30 meters from one survey collection point parallel to the trail edge by *pacing*. Even as the trail bends or terrain is uneven, the transect line should remain on the trail and parallel to the trail edge.
4. Flag (or mark) the start and end points of your transect line clearly. Record the start and end GPS coordinates for each belt transect on the data sheet.
5. For each 1 meter of the 30 meter transect line, measure (using a tape measure) 3 meters to the left (from the left trail edge) and 3 meters to the right (from the right trail edge), perpendicular to the transect line. For every 1 meter of the transect line that is surveyed, 3×1^2 meter sampling squares to the left of the left trail edge will be surveyed, and 3×1^2 meter sampling squares to the right of the right trail edge will be surveyed. This is clearly represented in *figure 1*.
6. Record the start and end times for surveying each belt transect.
7. For photo documentation: Take four photos for each transect.
 - a. Take a photo (landscape, 1x zoom) from the 0m mark looking up the transect line.
 - b. Take a photo (landscape, 1x zoom) from the 30m mark looking down the transect line.
 - c. Take a photo (landscape, 1x zoom) from the transect line looking left
 - d. Take a photo (landscape, 1x zoom) from the transect line looking right
8. Within each 1 x 1 meter sampling square, record on the data sheet which species (from the list of 23 target species) are present. A species is considered present if any portion of its aerial canopy is within the sampling square. Record all species that canopy overlaps the sampling square, including understory or groundcover species.
9. In the correlating cell on the data sheet record the plant species code for species present within sampling square (see species list), NS (for not surveyed), or a blank cell (for unrecorded).
10. Repeat steps 3-9 for belt transects two and three.
11. Input plant data and photo documentation into the web app. If recording on a physical data sheet, transcribe the data from the data sheet to the web app after the survey is complete.

Definitions

- *Belt transect*: A monitoring technique used to measure frequency, and presence/absence of plant or animal species along a transect line using quadrats.
- *Pacing*: Pacing will be used to measure the transect line. Pace out 30 meters, adding up the number of steps taken from measurement 0 to measurement 30m. This number of steps will be what you use to measure out 30 meters (from your survey collection point).

[illegible]

Figure 1: This table demonstrates the layout of the belt transect. The 30 meter transect line will be measured on the trail. The width of the transect belt will be measured 3 meters to the right of the right trail edge and 3 meters to the left of the left trail edge perpendicular to the trail. Each sampling square will be 1 meter by 1 meter.